These scripts read the current form of spreadsheet files, save them in their raw form, and then perform data cleaning and transformation to the target form, which is written to the database used by the REST API and files located on the shared drive. During data processing, processing logs, error logs, and a quality report are generated. The whole process is complemented by CI/CD pipelines and metadata describing the data product. All these elements constitute a data product—they are created and maintained by the data product development team.

The following sections describe the main elements of the data product's internal architecture.

Datasets

One of the most obvious components of a data product is the dataset. Datasets are what ultimately constitute the essence of a data product. As described in an earlier section, a dataset can be one or more related tables, a file, data in a stream, or successive versions of processed data. In many cases, the dataset will not physically reside where the other elements of the data product reside.

For example, the dataset may be a remote file whose size or legal aspects prevent copying. In particular, the latter may be the case when, because of licensing, the file can physically reside only within the boundaries of a geographical region. Another example is a dataset that is part of a database running as a cloud service.

Metadata

In the data mesh approach, metadata plays a particularly important role that allows many processes to be automated. Much of the metadata, especially descriptive and configuration metadata, will be part of the internal implementation of the data product—that is, its name, business description, dataset schema, people responsible, data-quality metrics, and available ports. Metadata in its physical form can be described, for example, as a JSON file. We'll look at metadata in a little more detail in section 5.5.2.

Code